Q1. The table below gives information about three fuels that can be used in cars.

✓ shows a substance is produced when the fuel burns.
X shows a substance is **not** produced when the fuel burns.

fuel	physical state	energy released, in kJ/kg	some of the substances produced when the fuel burns		
			carbon monoxide	sulphur dioxide	water
petrol	liquid	48 000	✓	✓	✓
hydrogen	gas	121 000	**X**	**X**	✓
ethanol (alcohol)	liquid	30 000	✓	**X**	✓

(a) Which fuel, in the table, releases the **least** energy per kilogram (kg)?

.....

1 mark

(b) Some scientists say that if hydrogen is burned as a fuel there will be less pollution.
From the information in the table, give **one** reason why there will be less pollution.

.....

.....

1 mark

(c) Which of the three **fuels** in the table can be compressed into a small container?

.....

1 mark

(d) Which gas in the air is needed for fuels to burn?
Tick the correct box.

carbon dioxide ☐

nitrogen ☐

oxygen ☐

water vapour ☐

1 mark

(e) Petrol and ethanol are both fuels. Petrol is made from oil.
Scientists say that oil could run out in 100 years.
In some countries people plant sugar cane and use it to make ethanol.

Sugar cane will **not** run out. Explain why.

.....

.....

1 mark
Maximum 5 marks

Mark schemes

Q1.

(a) ethanol **or** alcohol

if more than one box is ticked, award no mark

1 (L3)

(b) any **one** from

- burning hydrogen does not produce carbon monoxide

  *accept 'petrol **or** ethanol*
  ***or** alcohol produces carbon monoxide'*

- burning hydrogen does not produce sulphur dioxide

  accept 'petrol produces sulphur dioxide'

- burning hydrogen only produces water

- burning petrol causes acid rain

  *accept 'hydrogen **or** ethanol*
  ***or** alcohol does not cause acid rain'*

1 (L4)

(c) hydrogen

accept 'H_2"

accept 'gas'

1 (L4)

(d) oxygen ✓

if more than one box is ticked, award no mark

1 (L4)

(e) any **one** from

- it can be grown

  accept 'it does not take long to grow'

- it can be replanted

  accept 'it can be replaced'

- it is renewable

- it can be reproduced

  accept 'it produces seeds'

1 (L4)

[5]